Reconstruction, bare quantification, and the fourth reading*

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Abstract It is a near-consensus view in the semantic literature that the interaction between quantificational DPs and intensional operators does not give rise to *wide-scope opaque* readings of intensional sentences; namely, readings in which a DP in a sentence outscopes an intensional operator while its restrictor is evaluated opaquely, in the operator's domain of quantification. In a dissent, Szabó (2010, 2011) presents a set of intensional sentences—cutting across the three primary intensional environments: attitudes, modals and tense—argued to exemplify exactly this reading, and deduces from the apparent generality of the data that an unrestricted *bare quantification* mechanism is in charge of accounting for it. We argue that the phenomenon is not as general as claimed, illustrating that Szabó's wide-scope opaque reading arises for modal sentences only when the modal they contain is embedded in a relative clause, and the determiner under which the relative clause itself is embedded scopes over that modal, while the NP heading the relative clause is interpreted opaquely. We then argue that the problematic reading can thus be derived without non-standard scoping mechanisms, as the byproduct of *reconstruction* into intensional relative clauses. Since reconstruction effects are strictly constrained, our approach makes testable predictions regarding the distribution of Szabó's reading with modals, which are corroborated. While we focus only on Szabó's examples with modals, we conclude by outlining the kind of approach we believe will prove to be most fruitful in accounting for rest of the data.

Keywords: the fourth reading, wide scope opaque, relative clause, reconstruction, modals, bare quantification

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1 Introduction

Fodor (1970) famously claimed that a DP in an intensional environment may vary along two independent dimensions: It can take *wide* or *narrow* quantificational scope relative to the intensional operator in the sentence, and it can have its NP restrictor evaluated *opaquely*, in the worlds quantified over by that operator, or *transparently*, in the matrix world of evaluation. This gives rise to a four-way ambiguity, summarized in Table 1: On its *de dicto* reading, a DP takes narrow quantificational scope relative to an intensional operator, and its restrictor is evaluated opaquely. On its *de re* reading, the DP takes wide quantificational scope and its restrictor is evaluated transparently. The *third* reading combines narrow quantificational scope with transparent evaluation, and the fourth reading, henceforth **R4**, combines wide quantificational scope with opaque evaluation.

Sentence (1) is claimed in Fodor 1970 to demonstrate this ambiguity. Its *de dicto* reading in (1a), is true as long as buying *any* inexpensive coat meets Mary's desires. Its *de re* reading in (1b) is verified by there being a *specific* inexpensive coat Mary wants to buy, but she need not know that it is inexpensive, or, in fact, even a coat. The *third* reading in (1c) is true if buying each of several objects would satisfy Mary's desires, and all happen to be inexpensive coats, again, possibly unbeknownst to Mary. Finally, the R4 is true if there is a particular object that in all worlds that satisfy Mary's desires is an inexpensive coat that she buys.

(1) Mary wants to buy an inexpensive coat.

a. $\lambda w.\ \forall w' \in \text{DESIRE}_{Mary,w}\colon \exists x\colon \mathit{inexpensive\text{-}coat}_{w'}(x) \wedge \mathit{buy}_{w'}(\mathit{Mary},x)$ *(de dicto)*

b. $\lambda w.\ \exists x\colon \mathit{inexpensive\text{-}coat}_{w}(x) \wedge \forall w' \in \text{DESIRE}_{Mary,w}\colon \mathit{buy}_{w'}(\mathit{Mary},x)$ *(de re)*

c. $\lambda w.\ \forall w' \in \text{DESIRE}_{Mary,w}\colon \exists x\colon \mathit{inexpensive\text{-}coat}_{w}(x) \wedge \mathit{buy}_{w'}(\mathit{Mary},x)$ *(third)*

d. $\lambda w.\ \exists x\colon \forall w' \in \text{DESIRE}_{Mary,w}\colon \mathit{inexpensive\text{-}coat}_{w'}(x) \wedge \mathit{buy}_{w'}(\mathit{Mary},x)$ *(R4)*

On this approach, whether (1) asserts the existence of a specific coat depends on the determiner's scope relative to the attitude, whereas whether Mary is allowed to be ignorant about the fact that what she desires meets the description *inexpensive coat* depends on the restrictor's domain of evaluation.

The literature since Fodor 1970 has cast doubt on the empirical claim that (1) has the four readings above; in particular, the availability of the alleged R4 has been contested (cf. Keshet 2008, von Fintel & Heim 2011, Keshet & Schwarz 2019). If this reading is available, the sentence should be verified by scenarios in which Mary wants to buy a particular object, which is an inexpensive coat in Mary's DESIRE-worlds, though not necessarily in actuality. Assuming that Mary's DESIRE-worlds are constrained by what Mary *believes* (cf. Heim 1992), (1) is verified by the

	Transparent	Opaque
Wide-scope (*specific*)	*de re*	*fourth reading?*
Narrow-scope (*non-specific*)	*third reading*	*de dicto*

Table 1 DPs in intensional contexts

existence of an object that is actually expensive, which Mary wants to buy, thinking that it is inexpensive. As Keshet (2008: 30) points out, this is what (2a) means, but is not a possible interpretation of (1), as indicated by the fact that once pronominal reference is used to force wide scope for the DP over the attitude verb, as in (2b), we can no longer coherently deny that the coat is inexpensive.

(2) a. There's a coat that Mary wants to buy. She thinks it is inexpensive. But really, it is quite expensive.

b. Mary wants to buy an inexpensive coat. #But really, it is quite expensive.

The consensus that has taken shape since Fodor 1970 is thus that it is a mistake to assume that the two interpretative dimensions in Table 1—namely, wide *vs* narrow scope and transparent *vs* opaque evaluation—are truly independent of each other. Rather, wide scope for a DP relative to an intensional operator should block evaluation of its restrictor in the quantificational domain of the operator. In fact, as will be discussed in detail below, the semantic formalisms standardly used in the intensionality literature often have the non-existence of R4 built into them as a feature, as they reserve the possibility of opaque evaluation only to elements that are in the c-command domain of an intensional operator. On such an approach, as long as scope is determined structurally, wide-scope and opacity involve conflicting requirements, ruling out the possibility of R4.

As is usually the case, of course, not everyone subscribes to the consensus view. In particular, Szabó (2010, 2011) revived Fodor's empirical claim that R4 relative to attitude predicates exists and expanded on it, by arguing that the reading is not restricted to attitude reports and providing examples involving modals and tense operators as well. On this basis, Szabó also defends an expansive version of the main theoretical conclusion that Fodor draws from her empirical claim; namely, that opacity and wide scope should in fact be divorced one from the other and that we must posit a general mechanism dedicated to the generation of LFs in which a determiner outscopes an intensional operator that in turn determines the evaluation domain of its restrictor.

In this paper we scrutinize a particular example, given in (3), offered by Szabó as an illustration of R4 relative to a modal operator. Now, Szabó (2011) does not say much about this case, except that the truth of (3b) given the scenario in (3a)

indicates that it has a "split reading" in which the determiner *two* and the NP *judges* are evaluated in different positions with respect to the modal *could*. This, indeed, seems to be the case: Since only one person is stipulated to preside over Alex's court case in (3a), (3b) is false if the determiner scopes below the modal, together with its adjacent NP. And given that there are no *actual* judges involved in (3a), only potential ones, it is also false if the NP is evaluated above the modal together with the determiner. The sentence is only true in its respective context if it can be paraphrased with (3c), where the plural indefinite determiner *two* takes scope, and distributes,[1] over the modal *could*, while the NP *judges* is evaluated within the intensional environment. Hence, the DP seems to have R4 relative to the modal.

(3) a. **Context:** In Alex's district, judges are elected, not nominated. Two candidates are running to fill a single vacancy, neither of whom is or has ever been a judge. The winner will preside over a hearing in which Alex is a defendant. To make sure that her lawyer prepares for both possible outcomes, Alex says:

b. There are two judges we could face in this court.

c. There are two individuals in the actual worlds who could each be the judge we face in court.

(adapted from Szabó 2011: ex. 26)

This case is important, as it constitutes a crucial part of the motivation for Szabó's claim that R4 is a general phenomenon, cutting across the three primary intensional contexts—attitudes, modals, and tense. This claim, in turn, is taken to motivate a theory of intensionality and scope that fully divorces quantificational force from intensional status for almost any DP. In fact, Szabó posits a novel scope splitting operation, which allows a determiner to quantify unrestrictedly over the entire domain, with its apparent NP *restrictor*—obviously a misnomer on this approach—making its semantic contribution beneath some lower intensional operator within the nuclear scope of the unrestricted determiner. The resulting structure is schematized in (4), and is referred to—following Szabó—as *bare quantification*.

(4) [... Determiner ... [... Intensional operator ... [... NP ...]]]

In this present work, we will agree with Szabó that (3b) does, indeed, have a structure in which the determiner and its adjacent NP are evaluated in different positions relative to the modal. What we will contest, however, is that this lends support to a general bare quantification strategy in natural language. The gist of the argument is as follows: First, we argue that Szabó's proposed bare quantification mechanism overgenerates unattested cases of R4, while at the same time actually

1 More on the role of distributivity below.

undergenerating the observed R4 of (3b). These observations will serve to undermine the initial motivation for bare quantification.

We will then show that the availability of R4 for (3b) is contingent on its particular syntactic profile: The sentence contains a *relative clause* that embeds a modal operator, with the alleged R4 involving wide quantificational force for the determiner embedding the relative clause, and evaluation of the NP adjacent to it in the scope of the clause-internal modal. But, we will argue, if that is the case, no novel scope displacement strategy is required to derive the R4 of (3b) to begin with; relative clauses are known to exhibit *reconstruction effects* whereby the NP adjacent to the clause-embedding determiner is actually interpreted clause-internally.

Thus, when a relative clause contains a modal, reconstruction already mimics the effects of bare quantification, and we can reduce the R4 of (3b)—and R4 relative to modal operators more generally—to reconstruction. Since this approach takes the presence of a relative clause to be a precondition for R4 with modals, it avoids the overgeneration issues faced by bare quantification. And, since the availability of reconstruction into a relative clause is subject to stringent constraints, the approach will also be shown to further restrict the distribution of R4 in a way that correctly captures the empirical landscape; it is exactly those environments that rule out reconstruction into a relative clause, that also block R4 relative to modals.

From the perspective of the consensus view, this is a welcome result, as it reduces a challenging observation to familiar means. It is important, however, to note at the outset what this paper is *not* intended to achieve; that is, a full account of *all* of Szabó's examples of R4 that does not appeal to bare quantification. There are apparent cases of R4 relative to attitude predicates and tense operators that lack the relativization structure required to achieve R4 with modals, and thus cannot reduce to reconstruction effects. And while we briefly present some of these cases, and speculate on their nature in the conclusion, we leave a comprehensive treatment for future research. Our goals here are rather limited to:

(i) undermining the viability of Szabó's bare quantification mechanism;

(ii) motivating an account of the fragment of his data involving modals by appealing to independently-posited syntactic structures and within the confines of standard Montagovian semantics (Montague 1973);[2] and

(iii) briefly outlining the kind of approach we believe will be most fruitful in accounting for the remaining data, given the findings in this paper.

The paper is structured as follows: Section 2 presents Szabó's data, discusses its design, and highlights the challenge it poses to mainstream theories of intension-

2 In this respect, our approach contrasts with Moltmann's (2019, 2020) account of similar data within Fine's (2017) non-standard framework of *truth-maker semantics*. Engaging with this work is not our primary goal, but footnote 16 comments on the relationship between our account and this alternative.

ality. Section 3 spells out Szabó's bare quantification approach and the empirical challenges it faces, illustrating that the data used to exemplify R4 with modals is undergenerated by bare quantification, which also overgenerates unattested readings for related modal sentences. Section 4 then provides a reduction of R4 relative to modals to the reconstruction of NPs into intensional relative clauses. The predictions of this reductive approach are corroborated in Section 5, and a potential challenge is discussed in Section 6. Section 7 presents Szabó's remaining cases of R4, which are not amenable to the treatment we provide for the modal cases, and concludes by speculating on how they should be approached.

2 The fourth reading and the challenge that it poses

2.1 The main data point

As noted above, our focus in this paper will be Szabó's example of R4 relative to a modal operator, henceforth **R4-M**. The example, repeated in (5) for convenience, involves a sentence that seems true in a scenario that validates *only* its R4-M, suggesting that it must have a structure from which that reading, paraphrased in (5c), can be derived.[3]

(5) a. **Context:** In Alex's district, judges are elected, not nominated. Two candidates are running to fill a single vacancy, neither of whom is or has ever been a judge. The winner will preside over a hearing in which Alex is a defendant. To make sure that her lawyer prepares for both possible outcomes, Alex says:

b. There are two judges we could face in this court.

c. There are two individuals in the actual worlds who could each be the judge we face in court.

(adapted from Szabó 2011: ex. 26)

Note that, if possibility modals are taken to quantify existentially over worlds, Szabó's case of R4-M in (5) contains two existential quantifiers: the modal and a plural *indefinite* DP. However, two existential quantifiers are scopally-commutative; i.e., $[\exists x \exists y \phi(x,y)] \Leftrightarrow [\exists y \exists x \phi(x,y)]$ for any scope ϕ. Thus, if scopal-commutativity

3 Francez (2018) observed a phenomenon that at least superficially resembles Szabó's data, exemplified by an existential statement like "there can be three winners in this race", which can convey that three individuals might win the race. Here, too, the restrictor and determiner of a DP, *three winners* seem to scope in different places relative to a modal. However, such readings are subject to idiosyncratic restrictions (in particular, it is only available when relational nouns are involved), suggesting that they do not form a natural class with Szabó's data. We thus set this data aside and refer the reader to Francez 2018 for an account and to Nouwen 2022 for a recent discussion.

is assumed to apply when one of the existential quantifiers involved is an intensional operator, R4 of an existential DP relative to such an operator is predicted to be equivalent to its *de dicto* reading, and it is at least *prima facie* surprising that the two readings are distinguishable in the first place.

R4 is nevertheless distinguishable from *de dicto* in (5b), with the existential modal, presumably due to the presence of *distributivity*. The determiner in (5b), while existential, is also *plural* and therefore ambiguous between a reading on which it distributes over elements it dominates and one on which it does not. Distributivity introduces universal quantification over atoms of the pluralities witnessing the indefinite, and *that* quantification is not scopally-commutative with the existential intensional operator. Thus, without distributivity, R4 of, e.g., (5b) in (6b) is equivalent to its *de dicto* reading (6a), whereas its distributive reading (6c) is not.[4]

(6) a. $\lambda w.\ \exists w' \in \text{ACC}(w) : \exists_2 x : \mathit{judges}_{w'}(x) \wedge \mathit{face\text{-}in\text{-}court}_{w'}(x)$
(*de dicto*)

b. $\lambda w.\ \exists_2 x : \exists w' \in \text{ACC}(w) : \mathit{judges}_{w'}(x) \wedge \mathit{face\text{-}in\text{-}court}_{w'}(x)$
(*non-distributive R4*)

c. $\lambda w.\ \exists_2 x : \forall y \in \text{ATOMS}(x) : \exists w' \in \text{ACC}(w) : \mathit{judges}_{w'}(y) \wedge \mathit{face\text{-}in\text{-}court}_{w'}(y)$
(*distributive R4*)

(For any w, we take $\text{ACC}(w)$ to be the set of worlds accessible from w, and for any $n \in \mathbf{N}$, we use $\exists_n x$ as shorthand for $\exists x : |\text{ATOMS}(x)| = n$)

Thus, whenever an example below contains a plural indefinite, we focus exclusively on its distributive R4, inserting the silent operator DIST (7) at LF when this is required to make a truth-conditional difference. However, we discuss in Section 4 cases in which R4-M arises with determiners that are not indefinite at all.

(7) $[\![\text{DIST}]\!] = \lambda f_{\langle e,t\rangle} \lambda x_e.\ \forall y \in \text{ATOMS}(x) : f(y)$

2.2 The problem

The existence of R4 poses a challenge for a prominent approach to intensionality, exemplified by the *modified scope theory* (Keshet 2008), and the *world-pronouns*

4 As noted by a reviewer, whether commutativity applies when a modal is involved is subject to some controversy in the study of modal logic, and depends on whether we accept the *Barcan formula*, $\diamond \exists x : \phi x \rightarrow \exists x : \diamond \phi x$ and its converse, $\exists x : \diamond \phi x \rightarrow \diamond \exists x : \phi x$ (Barcan 1946), which are not universally accepted. Together, the formulas require that the domain of individuals remain constant as we move from the evaluation index to those indices that are accessible from it. This is trivially validated in the semantic formalism we take for granted, in which sentences are evaluated relative to a model that contains domains of worlds and individuals, modals are quantifiers over the former, and the latter does not change from world to world. Our choice of formalism can, of course, be questioned, but we think this debate is beyond the scope of the current paper, and thus circumvent it by focusing exclusively on the distributive construal of Szabó's data.

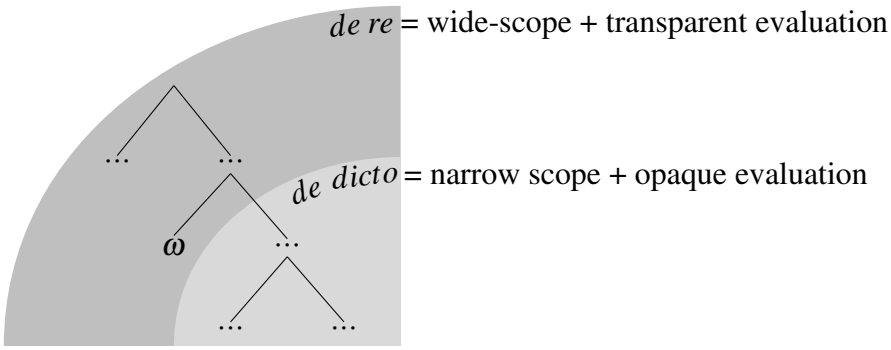


Figure 1 STI: wide-scope entails transparent evaluation; narrow-scope entails opaque evaluation

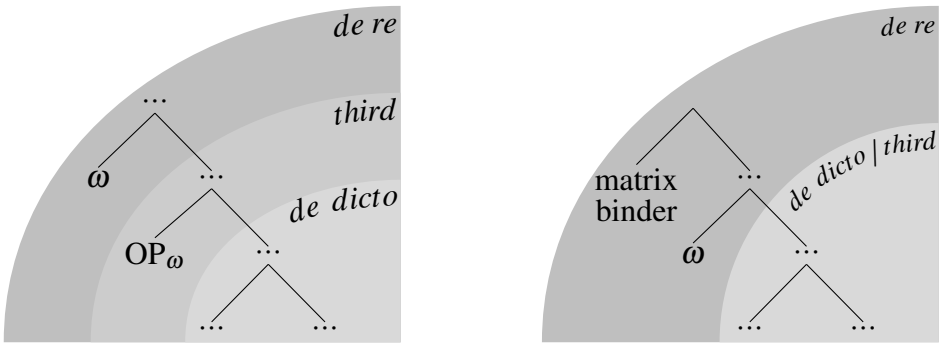
framework (Percus 2000). Developed in response to the shortcomings of the classical *scope theory of intensionality* (henceforth, STI; Montague 1973, Ladusaw 1977 a.o.), both retain STI's supposedly innocuous features. However, exactly those features cause the undergeneration of R4. Our discussion below is an abstract review of some basic semantic theory, which may seem superfluous to an experienced reader, but is included here nonetheless to underscore the problem and the range of theoretical possibilities available in addressing it.[5]

STI maintains that every world-dependent element below an intensional operator ω at LF is evaluated in the worlds quantified over by ω. Paired with a syntactic approach to scope, STI predicts a biconditional dependence between quantificational scope and intensional status: as shown in Figure 1, any element under ω will receive narrow-scope and opaque evaluation relative to ω, and any element taking scope above ω will receive transparent evaluation. Biconditional dependence undergenerates not only R4, but also the *third* reading, which requires narrow-scope, but transparent evaluation.

Keshet (2008, 2010b) modifies STI by divorcing the quantificational aspect of ω from the mechanism determining the evaluation domain of material in its scope. While being below ω at LF entails narrow-scope, it is not sufficient for opacity. DPs are opaque only if they reside below *another operator*, OP_ω, itself situated below ω at LF. In the resulting system of *split intensionality*, there is a region under ω where elements receive transparent evaluation; namely, the region between ω and OP_ω. As illustrated in Figure 2-a, syntactic position still fully determines a DP's intensional status, but STI's biconditional dependence becomes a conditional one: wide-scope entails transparency, but not vice versa.

The same conditional dependence is the result of Percus's (2000) more rad-

5 The discussion closely follows Keshet & Schwarz 2019.



(a) Split intensionality (b) World variables

Figure 2 Modifications of STI: wide-scope entails transparent evaluation while narrow-scope is compatible with transparent or opaque evaluation

ical departure from STI. In his system, world-dependent lexical items are functions whose input is a world-argument, saturated by silent object-language *world-variables* at LF. The restrictor of every DP is assumed to contain such a variable as its argument, which can be bound either by an intensional operator above it or by a binder posited at the matrix level, with the identity of the binder determining the intensional status of the DP. When a world-variable is bound by the matrix binder, the restrictor is transparent, and it is opaque when its variable is bound by the intensional operator above it. As shown in Figure 2-b, here, too, residing under ω does not determine a DP's intensional status, but being outside it does: ω cannot bind the world-pronoun of a DP above it, which must thus be evaluated transparently.

Both approaches, then, depart just enough from STI to generate the *third* reading, but not enough to allow opaque evaluation of wide-scope DPs, as is supposedly required for R4, presumably resulting in the undergeneration of any sentence that is judged true in an *exclusively*-R4-verifying scenario. Can we avoid this conclusion without rejecting the structural accounts of scope and binding? Perhaps, but our options seem limited. If Szabó's example truly involves a DP whose determiner takes wide-scope, that DP's restrictor cannot be interpreted opaquely unless, well, it does not serve to restrict the determiner it accompanies. Instead, this *pseudo*-restrictor must make its semantic contribution within the intensional environment that their determiner outscopes. We explore this next.

Before we continue, a technical note: For convenience, we will be using a system that posits object-language world variables. However, to avoid clutter we sometimes omit these variables and their binders from LF representations when all intensional variables are assumed to be bound locally, by the binder immediately dominating them.

3 Szabó's solution and its problems

3.1 *Bare quantification*

In introducing their Generalized Quantifier Theory (henceforth, GQT), Barwise & Cooper (1981) famously observe that quantificational determiners in natural language differ from predicate logic quantifiers ($\forall$, $\exists$) syntactically *and* semantically. Syntactically, they take a predicative complement, while predicate logic quantifiers combine with a formula. Semantically, their complement *restricts* their quantificational domain, whereas nothing restricts the domains of predicate logic quantifiers. Szabó argues that GQT got things wrong. While determiners always appear with predicates, they can quantify unrestrictedly like predicate logic quantifiers, with their complement predicate interpreted within their scope.

Szabó (2010) mediates the apparent mismatch between syntax, which requires determiners to take NP complements, and semantics, which allows them to quantify unrestrictedly, with a scope displacement operation, termed *quantifier splitting* (henceforth, QS) in (8), which allows the NP in the complement position of a raised DP to reconstruct into its pre-raising position without the DP's determiner. The determiner left in the derived position is restricted by an identity function, $\lambda x.x = x$ in (8), vacuously satisfying the syntactic requirement that determiners combine with predicates. The reconstructed NP is converted into an indexed definite description whose index is bound by the vacuously restricted determiner (cf. Fox 2003).[6]

(8) *Quantifier splitting (**QS**):*
$[_{\text{TP}}\,[_{\text{DP}}[D][_{\text{NP}}\,N]][[\lambda_\iota][_{\text{TP}}\ldots[_{\text{DP}}\,t_\iota]\ldots]]]$
$\rightarrow [_{\text{TP}}\,[_{\text{DP}}\,[D][\lambda x.x = x]]\,[[\lambda_\iota]\,[_{\text{TP}}\ldots[_{\text{DP}}\,\text{THE}\,[_{\text{NP}}\,N][\lambda y.y = x_\iota]]\,\ldots]]]$

By converting the reconstructed NP into an indexed definite description, QS triggers a presupposition in the scope of the vacuously restricted quantifier in the derived position. We will make the simplifying assumption that this presupposition is locally accommodated.[7] Effectively, then, when a DP consisting of a quantificational determiner and its NP restrictor undergoes QS, elements quantified over by the determiner are asserted to satisfy the function denoted by NP inside the determiner's scope. Thus, if the determiner's scope itself contains an operator that shifts the evaluation domain of its input, this operator will also shift the evaluation domain of the reconstructed NP, resulting in exactly the configuration required to derive R4.

Now, Szabó (2011) is not explicit about how QS is applied to derive the R4 of (5b), so the following contains some guesswork on our part. However, his assump-

6 Note that QS in (8) is a notational variant of Szabó's actual proposal that uses terms more familiar to linguists, which we use in Benbaji 2021.

7 In Benbaji 2021 we show that the arguments against QS cited below go through regardless of this.

tion that the truth of (5b) in the exclusively-R4-M-verifying scenario (5a) can be taken as evidence for a bare quantification strategy along the lines of (8), seems to imply that he views (5b) as containing a DP *two judges* that raises across the modal *could*. In that case, indeed, QS derives for (5b) the structure schematized in (9a), with the modal scoping between the determiner and a "restrictor" NP converted into an indexed definite. With the presupposition of that indexed definite locally accommodated, (9a) has (9b) as its truth-conditions. Since the modal resides between the original and derived positions of the raised DP, it shifts the evaluation domain of the reconstructed restrictor without scoping above the determiner, resulting in wide-scope for the determiner and opaque evaluation of the restrictor.

(9) a. $[_{\text{TP}}$ there are $[[[_{\text{DP}}$ [two][$\lambda x.x = x$]][[DIST][[λ_t][$_{\text{TP}}$ could
[we face $[_{\text{DP}}$ [the][[$\lambda y.y = x_t$][judges]]] in court]]]]]]]

b. $\lambda w.\ \exists_2 x\colon \forall y \in \text{ATOMS}(x)\colon \exists w' \in \text{ACC}(w)\colon$ we face the judge y in w'

3.2 Against *quantifier splitting*

There are two related reasons to reject QS as a possible scope displacement operation: First, it overgenerates, predicting R4s where they do not seem to be attested. In particular, R4-M seems restricted to particular relativization environments; namely, those in which the relevant determiner seems to take wide scope relative to a modal embedded in a relative clause (henceforth, RC) in its scope, headed by the NP that seems to be opaque relative to the RC-internal modal (cf. Benbaji 2021). This is illustrated by the modification of Szabó's modal case (5) in (10), as well as by several new examples in (11-13).[8]

8 This variety of examples is provided to show that the reading and its dependence on relativization can be demonstrated with different NPs. As a reviewer notes, some NPs seem to mysteriously resist the fourth reading, even within relativization structures that, it will be argued below, should suffice to give rise to R4-M. For instance, the following examples seem false in their respective, exclusively-R4-M-verifying scenarios:

(i) [Imagine an upcoming special election, next week, with two strong candidates for mayor. The winner will take office temporarily, until the general election next year. Tim is waiting to throw his hat in the ring in the general election:] *There are two mayors Tim could face in the general election next year.*

(ii) [Consider the following as a description of the plot of *Mamma Mia!*, where Sophie invites three of her mother's former lovers to her wedding, hoping to find out which of them is her father, who she hopes would walk her down the aisle:] *There are three fathers of the bride who could escort Sophie down the aisle.*

(iii) [Said in 2016, before the presidential elections:] *There are two U.S. presidents who could give the inaugural address next January (namely, Hilary Clinton and Donald Trump).*

We do not know why *mayor*, *father of the bride*, and *president* in (i-iii) differ from the NPs in

(10) a. **Context:** Alex is awaiting the judicial elections, as in (5a) only this time there are five non-judges as candidates. Alex believes only two stand a chance, and instructs her lawyer to prepare for either winning.

b. There are two judges who could end up presiding over our case. **true** in (10a)

c. Two judges could end up presiding over our case. **false** in (10a)

(11) a. **Context:** A building collapsed, with only the security guard on the night shift present inside. There are only two people who are ever on duty at night, but we are not sure which of them picked up the shift on the night of the collapse. Thus, for all we know, each of them *might* be trapped under the rubble, but we are certain that it is not the case that both are.

b. There are two victims who might be trapped under the rubble. **true** in (11a)

c. Two victims might be trapped under the rubble. **false** in (11a)

(12) a. **Context:** Three roommates were swabbed for a virus. The lab confirms that exactly one sample came back positive, but the labels were lost, so we do not know whose sample it was but we know that at most one of the roommates is a carrier. They are swabbed again. The carrier, once identified, will enter quarantine. The lab manager says:

b. There are three carriers who may have to be quarantined tomorrow. **true** in (12a)

c. Three carriers may have to be quarantined tomorrow. **false** in (12a)

(13) a. **Context:** A department had one tenure-track position to offer, and invited three graduating PhD students for job talks, each of whom the department was fully prepared to hire (the vote could have gone any of three ways). At the last minute, the dean canceled the search due to a hiring freeze.

(10-13). But since the phenomenon that (10-13) demonstrate seems robust enough to merit an explanation, we will set that question aside for the purposes of this paper. Note, also, that we can construct some marginal examples of R4-M even with an NP like "president". For instance, the following case does not seem as bad as the ones in (i-iii):

(iv) [Said in a rally by Donald Trump before the 2016 election:] *There are two presidents you all could have come January: An outsider who will fix this country—ME!—or an establishment type who is still busy deleting emails.*

Recent work (e.g., Zhang & Davidson 2024) suggests that the surface position of a noun plays a role in how difficult or easy it is for speakers to interpret it opaquely or transparently. This factor might be at play here, but of course, if that is the case, the question of why it does not seem to be a factor in the general case has to be addressed. I thank Florian Schwarz for pointing this out to me.

b. There are three professors who could have joined us in the fall. **true** in (13a)

c. Three professors could have joined us in the fall. **false** in (13a)

The (a)-contexts above make *only* the R4 of their respective (b-c)-examples true. However, while the (b)-examples, which contain RCs, seem felicitous given the exclusively-R4-verifying contexts, the (c)-examples, without relativization, seem false.[9] But QS generates a R4 for the (c)-examples, where it is widely assumed that the DP had raised over the modal, and can thus undergo the splitting operation, whose sole precondition is that it be preceded by such raising.

Note that we use the minimal pair in (10b-10c) rather than the one in (14a-14b) which contains Szabó's original sentence as one of its members, despite the fact that it gives rise to the same contrast as the one observed in (10), once the RC is eliminated from the sentence.

(14) a. There are two judges we could face in this court. **true** in (10a)

b. We could face two judges in this court. **false** in (10a)

This is because, one can plausibly argue that the contrast in (14a-14b) does not illustrate the central role of relativization in achieving R4-M. On the assumption that the existential DP and modal in these examples are scopally-commutative, discussed in Section 2.1, R4 of (14b) is only distinct from its *de dicto* reading when the plural DP scopes and *distributes* over the modal. But there is a known ban on or general dispreference for inverse-scope object-distributive readings, observed independently by, a.o., van der Does (1992), Steedman (2011), Križ & Maldonado (2018), which itself might explain the contrast in (14a-14b); i.e., perhaps QS is blocked independently in (14b) because the DP *two judges* cannot QR and distribute over the rest of the sentence. However, even if QS does not overgenerate a non-existent R4 for (14b) due to independent constraints on distributivity, it certainly overgenerates one for (10c) and the other (c)-examples above, and is thus *too expressive*.[10]

9 In a footnote, Szabó (2011) notes that according to his own judgment, a sentence like (10c) does have R4, though it is harder to tease apart than in cases with relativization, and an anonymous reviewer agrees. Several informants we consulted with judged the sentence as clearly out in an exclusively-R4-verifying scenario, and similar judgments about the general unavailability of R4 with modals are reported in Francez 2018. It is notable, in any case, that everyone in this debate seems to agree that it is at the very least much more difficult to achieve R4 without relativization than with it.

10 Santorio (2013) independently argued that QS overgenerates, focusing on unattested fourth readings of attitude reports with a proportional quantifier—elements like *most, one fourth, less than half*—in its prejacent. Traditionally, these quantifiers are assumed to compare the cardinality of two sets. Allowing one of those sets to correspond to the entire domain, QS derives a true reading of Santorio's (i-b) in the context of (i-a), contrary to fact: If the determiner *less than half* is vacuously restricted, while its NP complement *stockbrokers* is interpreted as part of its scope, (i-b) is wrongly predicted

This, in turn, leads us to the second problem with QS, which is—surprisingly—that QS is *not expressive enough* to generate a R4 for Szabó's original cases in (14a) or (10b). Once we notice the RC in the syntactic structure of these examples, we observe that QS actually *undergenerates* their observed R4 because, on a common assumption regarding the structure of RCs, discussed in more detail below, in (14a) the determiner above the RC and its adjacent NP—namely, *two* and *judges*—do not on their own constitute a DP that can undergo QS relative to the RC-internal modal *could*. But the splitting operation in QS must be preceded by movement of a quantificational DP, so if *two judges* does not constitute such a DP, the R4 of (14a) is undergenerated by QS.

These two arguments against Szabó's proposal are not of equal status. One is an argument against the very idea that natural language makes available a bare quantification strategy; if it did, we would observe fourth readings where we, in fact, do not. The other is an argument against a particular technical implementation of such a strategy; QS specifically predicts that we should not observe particular fourth readings which, in fact, we do. Given that Szabó, is not committed to the technicalities of QS, which are provided "for the sole purpose of illustration" Szabó (2011: 270), we should wonder whether the latter challenge—from undergeneration—can be defused.

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to have the R4 truth-conditions in (i-d), paraphrased in (i-c).

(i) a. **Context:** Sam believes all stockbrokers are rich, and that stockbrokers constitute a minority of existing things. Moreover, she thinks non-rich things vastly outnumber rich ones.

b. Sam believes less than half of the stockbrokers are rich. **false** in (i-a)

c. Sam believes that less than half of the domain's entities are rich stockbrokers. **true** in (i-a)

d. $\lambda w.\ \frac{|\{x \mid x=x\}|}{2} \geq |\{x \mid \forall w' \in \text{DOX}_{Sam,w} : x \text{ is a rich stockbroker in } w'\}|$

We are not sure whether this argument is conclusive, as there is some controversy regarding the scopal properties of proportional determiners, particularly those involving modified numerals like the one in (i-b). The controversy pertains to the question of whether these determiners can take wide scope at all (cf. van der Does 1992, Ruys 1992, Beghelli 1993, Schein 1993, Reinhart 1997, Winter 1997), and that argument has not been decided. Of course, if it turns out that wide quantificational scope for the DP in (i-b) is independently blocked—i.e., that "less than half of the stockbrokers" cannot QR in the first place—QS would not be expected to apply. There is some evidence that, indeed, the run-of-the-mill wide scope transparent reading of (i-b) is independently hard to achieve. Some speakers reject (i-b) in a scenario in which Sam thinks of certain actual stockbrokers that they are rich, but is either ignorant of the fact that they constitute less than half of the stockbrokers or thinks they constitute a majority of them. This suggests that wide scope might be blocked regardless of whether its restrictor is evaluated opaquely or transparently. We thank Kai von Fintel (p.c.) for this observation.

The argument in (10), we think, does not suffer from a similar objection, as nothing prevents a wide scope, *de re*, reading of (10c), but that sentence is nevertheless judged to be false in an exclusively-R4-M-verifying scenario.

It seems to us that defusing it without undermining the case for bare quantification is a difficult task. For instance, one may attempt to widen the applicability of QS somehow so that it applies even for the cases it currently undergenerates. But doing so can only worsen the overgeneration problem QS faces. Alternatively, one may concede that the undergenerated cases of apparent R4-M are not derived via the same mechanism that derives bare quantification elsewhere. In that case, if we do not want to worsen the overgeneration problem, we must also concede that these cases are not instances of bare quantification at all. But then the claim that R4 is a general phenomenon, cutting across intensional environments, is undermined, as we lack an example of the reading relative to modal operators.

We argue next that that the latter approach is correct; i.e., that since R4-M is contingent on relativization, its instances do not involve bare quantification. We will show that RCs already provide us with structures that can derive the attested cases of R4-M, and in all these cases, quantification *is* restricted, just not by the NP linearly succeeding the determiner; rather it is a reconstructed RC that restricts the determiner's domain of quantification.

4 Reducing R4-M to reconstruction into relative clauses

We have seen in (10-13) that R4 relative to modals, R4-M, is available only when the relevant modal is embedded inside a RC dominated by the DP with the apparent R4.[11] Next, we briefly review some structural properties argued to characterize

11 Szabó (2010) mentions the example in (i) that, prima facie, challenges this generalization, and a reviewer similarly suggests (ii).

(i) For her semantics seminar, Anna could write three papers. Now she has to decide which one to write.

(ii) You could build hundreds of different toys using the blocks in this LEGO set.

The sentence in (i), on its most natural interpretation, seems to convey that three things could be papers that Anna writes. Similarly, (ii) is judged true when hundreds of things could be toys built from the LEGO set, even if there are not enough blocks in the set to build hundreds of toys.

We think that the apparent R4-M of (i-ii) is due to their use of the creation verbs "write" and "build". R4-M seems unavailable for the variations in (i′-ii′), which—without a creation verb—have only a *de dicto* or *de re* reading. (ii′), for instance, seems committed to the claim that it is possible to sell hundreds of toys all formed from the discussed LEGO set. Note further, that in both cases, we can achieve R4-M by introducing a modal RC, as shown in (i″-ii″).

(i′) For her semantics seminar, Anna could read three papers.

(ii′) You could sell hundreds of different toys using the blocks in this LEGO set.

(i″) There are three papers Anna could read (depending on which of his ideas Ben plans to write about for his term paper).

(ii″) There are hundreds of different toys that you could sell using the blocks in this LEGO set.

RCs, and illustrate how several such properties by themselves derive the scopal configuration for which Szabó posits QS. As an independent mechanism, then, QS is *superfluous* in the derivation of R4-M. We then provide evidence that R4-M is indeed restricted to DPs with RCs that exhibit exactly those structural properties that derive R4-M.

4.1 Raising relative clauses

Brame (1968) first suggested that RCs have LFs in which the head NP originates RC-internally (see also Schachter 1973, Vergnaud 1974, and Kayne 1994). These raising LFs are posited to derive the observation that some grammatical RCs comply with independent syntactic conditions *only* if their heads originate in RC-internal positions. For instance, the head-NP in (15a) contains an anaphor which satisfies Binding Condition A only when bound by a binder that resides inside the RC. The anaphor must have therefore originated below its binder, and binding conditions must have been "checked" before it raised to its RC-external position. Relatedly, in (15b) the head forms an idiom with an embedded VP. Assuming that the verbal argument in VP idioms must reside in the verb's complement position at LF, this indicates that the RC-head had raised from VP's complement position. Crucially, then, the examples in (15) are grammatical because they can be parsed as if the RC-head is interpreted in its pre-raising position.

(15) RCs *only* compatible with a raising structure:

a. Condition A: $[_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \ldots\ \text{anaphor}_i\ \ldots]_j\ [_{\text{RC}}\ \ldots\ \text{R-expression}_i\ \ldots\ \text{t}_j]]$

i. The rumor about himself$_i$ that John$_i$ denied is unflattering.

ii. The interest in each other$_i$ that John and Mary$_i$ showed was fleeting.

b. Idiom chunks: $[_{\text{DP}}\ \text{D}\ [_{\text{NP}}\ \text{idiom part}]_j\ [_{\text{RC}}\ \ldots\ \text{idiomatic environment}\ \text{t}_j\ \ldots]]$

i. Mary could not ignore the tabs that John kept on his boss.

ii. The beans that John spilled shocked everyone.

The two properties in (16) often attributed to such *raising* relative clauses (henceforth, RRCs) will be taken for granted in our discussion. Both assumptions have

This suggests that the apparent R4-M in (i-ii) should not be derived via a general mechanism dedicated to the derivation of R4-M, but rather from the semantics of creation verbs. Notably, on a prominent approach, such verbs are themselves treated as intensional operators, so that a sentence like 'John built a toy' asserts that a would-be toy has been instantiated in actuality by John (cf. Bennett 1977). But then, (i-ii) might just be true on their *de re* readings in contexts that would seem to verify their alleged R4-M: In (i) there are three *actual* would-be papers that Alex could instantiate by writing, and in (ii) there are hundreds of actual would-be toys that could each be instantiated using the LEGO set. We can thus tentatively conclude that (i-ii) should not be taken to bear on our theory of R4-M.

been defended on independent grounds: (16a) is required for type reasons (Heim & Kratzer 1998: 96-8), and to account for the behavior of RRCs under presuppositional determiners (Partee 1975: 230-1).[12] More controversially, (16b) is motivated by various syntactic arguments, including the data in (15): Reconstruction of the head into the RCs in (15) is required to explain their apparent compatibility with syntactic well-formedness conditions.[13] Note, also, that the property in (16a) is what prevents QS from deriving R4-M for Szabó's example in (14a): The determiner and adjacent NP in that example originate in different structural positions, and thus cannot constitute the input to QS to begin with.[14]

(16) a. A determiner under which a RRC is embedded originates RC-externally, unlike the head NP. Thus, RRC-embedding DPs have the following structure: $[_{DP}\,[D][[\underset{\uparrow}{NP}][_{RC}\,\ldots\,[\underset{\uparrow}{\cancel{NP}}]\,\ldots\,]]]$.

b. The head NP, which originates RC-internally and raises to its surface position, can *reconstruct* into its pre-raising position at LF.

Bassi & Rasin (2018) propose the three-step derivation in (17) for RRCs with a reconstructed head NP, building on Heim 2019 (in circulation since 2012).

(17) a. *Merge copy of NP ("raising")*:
$[_{RC}\,\ldots\,[_{DP}[D][_{NP}\,N]]] \rightarrow [[_{NP}\,\underset{\downarrow}{N}][[\lambda_i][_{RC}\,\ldots\,[_{DP}[D][_{NP}\,\underset{\uparrow}{N}]]]]]$

b. *Trace conversion:*

i. $[[_{NP}\,N][[\lambda_i][_{RC}\,\ldots\,[_{DP}[D][_{NP}\,N]]]]$
$\rightarrow [[_{NP}\,N][[\lambda_i][_{RC}\,\ldots\,[_{DP}[D][_{NP}\,[N][\lambda y.y = x_i]]]]]]$

ii. $\rightarrow [[_{NP}\,N][[\lambda_i][_{RC}\,\ldots\,[_{DP}[\text{THE}][_{NP}\,[N][\lambda y.y = x_i]]]]]]$

c. *Higher copy LF-deletion ("reconstruction"):*
$[[_{NP}\,N][[\lambda_i][_{RC}\,\ldots\,[_{DP}[\text{THE}][_{NP}\,[N][\lambda y.y = x_i]]]]]]$
$\rightarrow [[\lambda_i][_{RC}\,\ldots\,[_{DP}[\text{THE}][_{NP}\,[N][\lambda y.y = x_i]]]]]$

The proposal treats raising as the merging of copies, and reconstruction as the deletion of certain copies at LF. Step (17a) thus involves merging a copy of the NP

12 See also Heim 1987: 36-37 and Bhatt 2002: 70-71 for further arguments. In particular, Heim shows that RCs under a strong determiner in the sense of Milsark 1974 can have their head originate in the post-copular position of a *there*-construction embedded within the RC. Since strong determiners are barred from this position, the determiner itself could not have originated RC-internally.

13 See Bhatt 2002 for a critical assessment of these arguments, and for additional ones. The controversy regarding syntactic reconstruction is beyond our focus, though we briefly remark in footnote 25 on how R4 might bear on it.

14 Moreover, while (16a) is a property unique to *raising* RCs, QS would also fail to apply in the desired way if the RC in (14a) were not analyzed as involving raising. On a particular version of the non-raising structure known as the "matching" construal, the determiner and NP do form a DP on their own, but that DP originates RC-externally as a whole, and QS would therefore never be able to place the NP in a RC-internal position below the modal, as would be required to achieve R4-M.

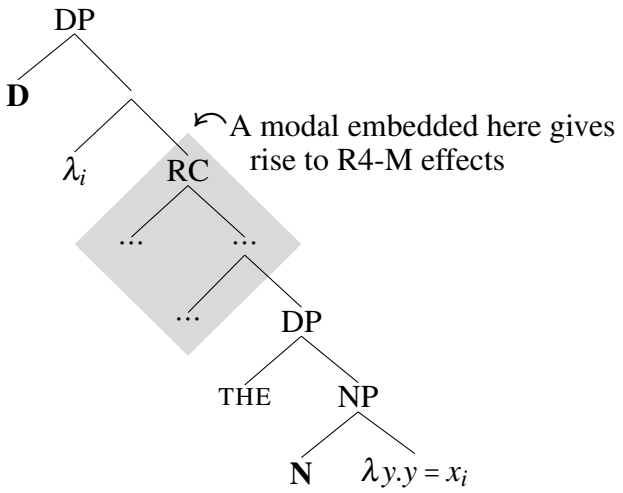


Figure 3 Structure of DPs embedding reconstructed RRCs

restrictor of a RC-internal DP in a RC-external position, together with a λ-binder directly below it. Next, the RC-internal DP whose restrictor has moved undergoes *trace conversion* (17b); i.e., it is converted into an indexed definite description whose index is bound by the λ-binder introduced by raising (Fox 2003). Finally, the higher *pronounced* copy of NP is ignored for interpretative purposes (17c). Note that, for brevity, we will sometimes use the long-form $[_{\text{DP}}[\text{THE}][_{\text{NP}}[\text{N}][\lambda y.y{=}x_i]]]$ to represent trace-converted DPs, and sometimes its shorter counterpart $[_{\text{DP}}[\text{THE}_i][\text{NP}]]$.

The properties in (16), together with the account in (17), derive the structure in Figure 3 for DPs embedding a reconstructed RRC. Crucially, the structural relationship between the RC-external determiner and the RC-internal NP, both boldfaced in the Figure, is exactly the relationship between the determiner and the nominal complement of a DP that has undergone Szabó's QS (8). When an intensional operator is situated between a DP's original and derived positions, QS generates R4. However, given our observation that R4-M in particular seems restricted to the kind of structures schematized in Figure 3, QS is unnecessary: Whenever a modal is part of the shaded region in the figure, R4-M comes for free, simply in virtue of reconstruction.

Less abstractly, Szabó's example of R4-M (18) takes the form of a *there*-sentence (Milsark 1974) with a numeral in its post-copular position. For ease of exposition, we treat a numeral *n* as an existential quantifier over *n*-sized pluralities that can distribute over its restrictor argument. This is implemented below by having a distributivity operator DIST (7) adjoin the restrictor of *n*. The *there*-sentence as a whole is then assumed to behave as if its post-copular DP has vacuous scope; i.e., it asserts the existence of a *n*-sized plurality whose atoms *each* make the embedded NP true,

as in (19).[15]

(18) There are two judges we could face in court.

(19) $[\![$[There are $[_{\text{DP}}$ $[n$ [DIST NP]]]]$]\!] = \lambda w.\ \exists_n x : \forall y \in \text{ATOMS}(x) : [\![\text{NP}]\!](y)(w)$

Following the recipe in (17), we construct the LF structure in (20a) for the underlined NP in (18). This LF derives the set of individuals for whom there is an accessible world in which they are the judge Alex faces in court (20b). The sentence thus asserts that there are two individuals, each of whom could be the judge Alex faces in court. This, of course, is the R4 paraphrase of (18).

(20) a. $[[\lambda_i][_{\text{RC}}$ $[_{\text{TP}}$ [could] $[_{v\text{P}}$ we face $[_{\text{DP}}[\text{THE}_i][_{\text{NP}}$ judge]] in court]]]]

b. $[\![$(20a)$]\!] = \lambda x \lambda w.\ \exists w' \in \text{ACC}(w)$:
we face in court in w' the unique y who is a judge in w' s.t. $x = y$

Note that on this analysis, both the choice of determiner in (18) and its *there*-sentence structure, are entirely orthogonal to the availability of R4-M. All that is required for R4-M is that the determiner, *whichever it is*, embed a RRC with a modal outscoping the original position of its head-NP. That this is indeed the case is illustrated by the examples in (21), which do not have the structure of an existential *there*-sentence and whose determiners are not numerals but are nevertheless able to quantify over potential, rather than actual judges.

(21) a. **Context:** Alex's lawyer obtains a list of potential judicial election candidates and presents it to Alex.

b. If the list exhausts all potential candidates, the lawyer can felicitously utter: "*Every judge we could face in court is on this list.*"

c. If Alex thinks the list does not in fact contain any potential candidates, she can felicitously utter: "*No judge we could face in court is on this list.*"

d. If the list contains some but not all potential candidates, the lawyer can felicitously utter: "*A judge we could face in court is on this list.*"

e. If the lawyer is certain that the eventual winner is listed in the document, they can felicitously utter: "*The judge we will face in court is on this list.*"

(adapted from Benbaji 2021: ex. 22)

15 Distributivity is necessary, though not *this* implementation (cf. Section 2.1). Both our treatment of numerals as determiners, and our assumptions regarding the meaning of Milsark's *there*-construction, are not essential, but are rather useful simplifications. We could have easily assumed, with, e.g., Keshet (2010a) that *there*-sentences involve existential closure over a predicative element, and that a numeral n in the post-copular position is a predicative element that restricts this existential quantification which can distribute over material below it. We only crucially assume that a numeral may originate RC-externally even when an NP below it heads a RC.

5 Reconstruction as a precondition for R4-M

Let us take stock: In previous sections we argue that bare quantification should not be generally available, given its overgeneration problem, and is also not needed to derive R4-M, given that R4-M is restricted to relativization environments and can thus, in principle, be derived via reconstruction. We now want to justify a stronger thesis; namely, that R4-M *must* be derived as a side-effect of reconstruction.

If this is the case, we predict the reading to be blocked whenever reconstruction is ruled out independently. The literature provides several diagnostics for the absence of reconstruction, which—we show next—correlate in the expected way with the unavailability of R4-M.[16] It is important to note that this next part of the paper no longer takes Szabó's argument for bare quantification as its foil; rather, it can be viewed as a "sanity check" for the idea that reconstruction in particular is in charge of deriving R4-M when this reading is attested.

5.1 Resumption

Sichel (2014) shows that Hebrew direct-object resumptive pronouns (hereafter, RPs) in RC-internal positions block reconstruction. This is illustrated by the illicitness of such RPs in environments that *require* reconstruction. In (22), for instance, reconstruction is required for an anaphor to satisfy Condition A. In (23), it is required to preserve the integrity of the idiom *sew a case*, meaning *to falsely implicate in a crime*. Both environments disallow RPs.

(22) ha-ʃmu'a al acmo$_i$ ʃe-Dani$_i$ hikxiʃ (*ota) hufca al yeday
the-rumor about himself that-Dani denied (*it.RP) was.spread on hands.of
Rani
Rani
'The rumor about himself that Danny denied was spread by Rani.'
(Sichel 2014: ex. 4)

16 The ability to capture the correlated distribution between R4-M and reconstruction constitutes, we think, a point in favor of our proposal compared to an alternative truth-maker semantic account, due to Moltmann (2019, 2020), who derives Szabó's data by positing a domain of *variable-objects*, each associated with their own concept-like function from *minimal-situations* to *manifestations-in-a-situation*. While these variable-objects may have different manifestations in different situations, they are nevertheless *objects* in the ontology, rather than functions. As we understand it, on this proposal Szabó's example of R4-M (18) is rendered true by there being in our domain two variable-objects that each have a manifestation as a judge Alex faces in some possible court situation. It seems to us that such a system can explain the correlation of R4-M and reconstruction only by positing distinct constraints on variable-object-denoting expressions that tie them to the structure of the sentences in which they appear, which would have to involve relativization and reconstruction. Here, we show that reconstruction itself can do the job without positing non-standard ontological domains.

(23) ha-tik ʃe-tafru (#oto) la-sar haya kaʃur le-nadlan
the-case that-they.sewed (#it.RP) for.the-minister was related to-real.estate
'The case that they pinned on the minister was related to real estate.'
(Sichel 2014: ex. 5a)

In Benbaji 2021, we illustrate that direct-object RPs block R4-M, as predicted by the reductive approach. The Hebrew translation of Szabó's example with a RP in (24) is infelicitous in Szabó's exclusively-R4-M-verifying scenario (5a), as it seems to commit one to the existence of *actual* judges.

(24) yeʃ ʃney ʃoftim ʃe-ʔanu ʔalulim lifgoʃ (#ʔotam) bevet hamiʃpat
there.are two judges that-we could meet them.RP in.house the.law
'There are two judges we could face in court.' (Benbaji 2021: ex. 24)

However, the reductive approach makes another, previously unexplored prediction regarding resumption: While optional direct-object RPs block reconstruction, obligatory *indirect*-object RPs allow it. Thus, Condition A is satisfied in (25), and idiomatic interpretation of the phrase *climb on/descend from a tall tree* (which, roughly, mean *to insist/relinquish an entrenched position*) is available in (26), despite their RPs.

(25) ha-ʃmu'a al acmo$_i$ ʃe-Dani$_i$ xaʃaʃ *(mimena) hufca al
the-rumor about himself that-Dani feared from.it.RP was.spread on
yeday Rani
hands.of Rani
'The rumor about himself that Danny feared was spread by Rani.'
(Sichel 2014: ex. 9)

(26) a. ha-ec ha-gavoha ʃe-hu tipes *(alav)
the-tree the-tall that-he climbed on.it.RP
'The position that he was stubbornly entrenched in'

b. ha-ec ha-gavoha ʃe-hu yarad *(mimeno)
the-tree the-tall that-he descended from.it.RP
'The entrenched position that he relinquished'
(adapted from Sichel 2014: ex. 10a-b)

Strikingly, RP-position correlates with the un/availability of R4-M in the way one would expect if R4-M is a byproduct of reconstruction. The variant of (24) in (27), in which the head NP *judges* raises from an indirect- rather than direct-object position of the RC-internal verb, is felicitous in the exclusively-R4-M-verifying scenario, despite the RP.

(27) yeʃ ʃney ʃoftim ʃe-ha-goral ʃeli bevet hamiʃpat ʔalul lihyot
there.are two judges that-the-fate mine in.house the.law could be
taluy *(bahem)
dependent on.them.RP
'There are two judges my fate in court could depend on.'

This is expected on the reductive approach as, given that reconstruction is *not* blocked by indirect-object RPs, nothing should block the effect of R4-M.

5.2 Extraposition

Reconstruction is also argued to be blocked for RC adjuncts extraposed over a right-adjoined adverb. Hulsey & Sauerland (2006) illustrate this by attempting to extrapose RCs that comply with independent syntactic constraints *only* if their head reconstructs into a RC-internal position. The resulting ungrammaticality or infelicity in (28) suggests that reconstruction, required for the satisfaction of Condition A in (28a) and to achieve an idiomatic interpretation of (28b), is blocked by extraposition.

(28) a. * I saw the portrait of himself$_i$ yesterday that John$_i$ likes.[17]

b. # Mary could not ignore the tabs yesterday that John was keeping on her.

As predicted, R4-M is indeed blocked when the modal-embedding RC is extraposed. In the exclusively R4-M-verifying scenario (29a), the example without extraposition seems true (29b), whereas its counterpart with it in (29c) seems false.

(29) a. **Context:** Alex is again awaiting the results of the judicial elections, whose candidates are again all non-judges, to see who will preside over her court case. She is a well-known politician and is encouraged by her lawyer to use her political platform to praise all judicial candidates, so that whoever wins will judge her favorably. She says:

b. I praised every judge that we could face in court. **true** in (29a)

c. I praised every judge yesterday that we could face in court. **false** in (29a)

(adapted from Benbaji 2021: ex. 27)

17 This is Hulsey & Sauerland's example. Some speakers may judge the sentence to be better than predicted if extraposition indeed blocks reconstruction, given that animate anaphors are sometimes exempt from binding condition A (cf. Charnavel & Sportiche 2016, Charnavel 2019). To avoid this confound, we may consider the contrast in (i), where an inanimate anaphor is used.

(i) a. I noticed the citation of itself$_i$ that the paper$_i$ mentions in a footnote.

b. * I noticed the citation of itself$_i$ yesterday that the paper$_i$ mentions in a footnote.

I thank Isabelle Charnavel and Dominique Sportiche (p.c.) for this point.

This falls out from the reductive account, of course, if extraposition blocks reconstruction, which is a precondition for R4-M.

5.3 Adjectival modification of the head noun

Next, we explore a consequence of the reconstruction account of R4-M, when combined with a particular hypothesis regarding the way reconstruction in raising RCs interacts with adjectival modification, and an independently motivated constraint on the evaluation world of modifiers. Unlike the previous predictions we tested, this one requires a more involved introduction.

The hypothesis that we will explore here states that raising RCs with adjectival modification, i.e., those with the surface form in (30), may be mapped only to a restricted set of LF structures in which the adjective and the noun are interpreted in the same position. This means that when independent considerations force an adjectival modifier to be interpreted RC-externally, the head noun will not be able to reconstruct, ruling out the possibility of R4-M.

(30) Det Adjective Noun RC

More precisely, the hypothesis states that the range of possible structures one could assign to a DP with the surface form in (30) is as follows: The adjective can originate with the noun that it modifies inside the RC, and raise with it to be interpreted RC-externally (31a), or reconstruct with it to be interpreted RC-internally (31b). The adjective may also merge RC-externally, if the noun itself is interpreted in its derived, RC-external position (31d). The hypothesis rules out structures in which the adjective raises with the head noun, but reconstructs without it (31c), and those in which the adjective originates RC-externally, but the head noun reconstructs to be interpreted RC-internally (31e).

(31) a. $[_{DP}\,[D][[_{NP}\,\underset{\uparrow}{AP}\ N][_{RC}\ldots[_{NP}\,\underset{|}{\sout{AP\ N}}]\ldots]]]$

b. $[_{DP}\,[D][[_{NP}\,\underset{|}{\sout{AP\ N}}][_{RC}\ldots[_{NP}\,\underset{\uparrow}{AP}\ N]\ldots]]]$

c. * $[_{DP}\,[D][[_{NP}\,AP\ \underset{|}{\sout{N}}][_{RC}\ldots[_{NP}\,\underset{|}{\sout{AP}}\ N]\ldots]]]$

d. $[_{DP}\,[D][[AP]\,[[_{NP}\,\underset{\uparrow}{N}][_{RC}\ldots[_{NP}\,\underset{|}{\sout{N}}]\ldots]]]]$

e. * $[_{DP}\,[D][[AP]\,[[_{NP}\,\underset{|}{\sout{N}}][_{RC}\ldots[_{NP}\,\underset{\uparrow}{N}]\ldots]]]]$

This hypothesis—ruling out structures with partial reconstruction (31c) as well as structures that mimic the interpretative effects of partial reconstruction (31e)—seems to us quite intuitive. However, our goal here is not to motivate a comprehensive account that derives it.[18] Rather, what we will show is that, when paired

18 For completeness, this footnote offers a potential account; namely, a set of selection and labeling

with the reconstruction view of R4-M and an independently motivated generalization about the evaluation world of modifiers (to be introduced next), this hypothesis makes exactly the right predictions with respect to the distribution of R4-M; i.e., that it is in fact the case that interpreting an adjective RC-externally blocks R4-M with respect to the head noun.

Keshet (2010a) shows that two predicates interpreted intersectively must have the same intensional status, motivating the generalization in (32). This is illustrated with the infelicity of the sentences in (33), which involve NPs consisting of a noun and a modifier that contradict each other, and should have been felicitous if the contradicting noun and modifier could have different evaluation worlds.[19]

(32) **The intersective predicate generalization:** Two predicates interpreted intersectively may not be evaluated at different worlds from one another.

(33) a. # Mary thinks the married bachelor is messy.

b. # Mary thinks the professor in college is too young to teach.

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principles that conspire to give us the picture in (31). Note that this is done for illustrative purposes only, as the literature on the interaction between labeling and selection in the context of RC structure is quite vast, and we cannot do justice to it within the confines of the present discussion (see, for instance Cecchetto & Donati 2015, Salzmann 2017 for some recent work and comprehensive bibliographies).

Consider the following principles:

(i) Determiners select either for NPs, or for constituents labeled with whatever label is associated with RCs, let us assume that this is CP.

(ii) the label on the constituent containing the RC and its head NP depends on whether the head reconstructs: with reconstruction, that constituent inherits the label of the RC—namely, CP—but without reconstruction, it inherits the label of the head NP.

(iii) NPs can have APs as their adjuncts, but crucially CPs cannot.

In (31a), the adjective is an adjunct of an NP, satisfying (iii), the modified NP does not reconstruct and so the constituent that dominates it inherits its NP label according to (ii), and so the determiner is merged with an NP, satisfying (i). In (31b), since the modified NP reconstructs, the constituent that combines with the determiner inherits the label of the RC, namely CP, and this is also permitted by (i), resulting in an acceptable structure. In (31d) the adjective does not originate as a modifier of the head NP in its RC-internal position. However, the structure is acceptable because the raising NP does not reconstruct, so the constituent dominating it in its derived position is itself an NP that can have an AP adjunct. The ungrammatical structures are ruled out as follows: In (31e) the adjective originates RC-externally, but the head noun reconstructs to its original, RC-internal merging position. Thus, (ii) dictates that the constituent that AP is merged as an adjunct of inherits the RC's CP label, violating (iii). The structure in (31c) is not ruled out by the principles in (i-iii), but rather by the reconstruction algorithm in (17); reconstruction is the deletion of the higher copy of the moved phrase, and the moved phrase in (31c) contains the adjective *and* the noun, which can thus only be deleted together.

19 See Keshet 2010a for further data in support of the generalization, and von Fintel & Fox & Iatridou 2014 for a critical discussion.

(Keshet 2010a: ex. 19)

With this in mind, consider a sentence ϕ containing a DP with the surface form in (30), in which the adjective involved is intersective,[20] and the RC involved contains a modal. And assume further that ϕ is uttered in a scenario that verifies it only if the noun is evaluated opaquely relative to the RC-internal modal while the intersective modifier is evaluated transparently. In other words, the background context verifies only a *de re* reading of the modifier part of the RC-head, but only the R4-M of the noun that it modifies.

The structures for the RC-embedding DP that could, in principle, derive such a reading for ϕ, are all ruled out by the independent assumptions introduced above. One such structure has the head noun raise with its adjectival modifier from a RC-internal position, but reconstruct without it (34a). Another has the modifier originate RC-externally, but has the head noun reconstruct (34b). These are ruled out by the hypothesis that structures with partial reconstruction and those mimicking them are ungrammatical. A final structure that would derive such a reading involves raising and reconstruction of the entire modified NP, but evaluation of the noun and the adjective in different evaluation worlds (34c). This structure, in turn, is ruled out by Keshet's in (32).

(34) a. * $[\lambda w \ldots [_{\text{DP}}\, [\text{D}][[_{\text{NP}}\, \text{AP}_{w}\, \sout{\text{N}}][_{\text{RC}} \ldots [\text{Modal}\ \lambda w' \ldots\ [_{\text{NP}}\, \sout{\text{AP}}\, \text{N}_{w'}] \ldots]]]]]$

b. * $[\lambda w \ldots [_{\text{DP}}\, [\text{D}][[\text{AP}_{w}]\, [[_{\text{NP}}\, \sout{\text{N}}][_{\text{RC}} \ldots [\text{Modal}\ \lambda w' \ldots\ [_{\text{NP}}\, \text{N}_{w'}] \ldots]]]]]]$

c. * $[\lambda w \ldots [_{\text{DP}}\, [\text{D}][[_{\text{NP}}\, \sout{\text{AP N}}][_{\text{RC}} \ldots [\text{Modal}\ \lambda w' \ldots\ [_{\text{NP}}\, \text{AP}_{w}\, \text{N}_{w'}] \ldots]]]]]$

We thus predict ϕ to be judged as false in a scenario that renders true only a *de re* reading of the modifier part of the RC-head, but only the R4-M of the noun that it modifies, due to a combination of Keshet's generalization and the hypothesis

20 Note that on a particular view of adjectival modification, the class of adjectives that count as intersective for our purposes is larger than might seem at first. *Gray* is intersective: Being a *gray elephant*, entails being gray and being an elephant. *Small* seems non-intersective: being a *small elephant*, does not entail being small and being an elephant; if it did, being a small elephant would be wrongly predicted to entail being small. Rather, *small* belongs to a class of adjectives often termed *subsective* (Partee 1995, Kennedy 2012). However, as Heim & Kratzer (1998) and Larson (1998) suggest, if subsective adjectives are assumed to come with a contextually supplied comparison class, then they, too, become intersective: If *small* is taken to be true of an individual x iff x is small given a contextually salient comparison class C, then being a *small elephant* does entail being small (given the comparison class) and being an elephant. On this view, Keshet's generalization should apply widely, even to so-called subsective adjectives.

spelled out above.[21]

The prediction seems to be borne out. Consider, for instance, the sentence in (35b) relative to the context in (35a).

(35) a. **Context:** Alex is accused of stealing from the rich to give to the poor. An election will determine which of ten judicial candidates (none of whom is or has ever been a judge) will preside over her court case, and Alex wants her lawyer to frequent social gatherings that those candidates attend, in an attempt to gain their sympathy. However, she thinks that candidates who are themselves among the rich will never be sympathetic to her cause, and wants her lawyer to focus on the poor candidates, even though she knows that whoever ends up winning—rich or poor—will not be poor when deciding her case, given the high salary judges are paid. She also thinks that only five of the ten candidates might actually win the election, three of which are poor. She finds out that these three candidates will be attending a party, and tells her lawyer: You must attend that party…

b. Every poor judge that we could face in court will be there. **false** in (35a)

c. Every poor candidate that we could face as the judge in court will be there. **true** in (35a)

The sentence in (35b) is quite odd in the scenario above, exactly because it can be true in that scenario only if the modifier in the RC head is transparent, while the noun itself is reconstructed to achieve R4-M: The universal quantifier must quantify over judicial candidates, not over actual judges, who are poor in actuality, but not in those worlds in which they win the judicial election. Such a reading for (35b), paraphrasable as in (35c), requires an illicit structure, hence the oddity of (35b) in the above scenario.

5.4 Superlative modifiers

Bhatt (2002) observed that a RC whose head contains a superlative modifier is

21 As a reviewer points out, Keshet (2010a) does not explicitly discuss violations of (32) in cases that require reconstruction. However, we can show that the generalization regulates such cases as well. For instance, the example in (i) is as infelicitous as those in (33), even when the indefinite DP *a married bachelor* reconstructs under the raising predicate *appear/seem* for scope reasons. For instance, suppose we are allowed to peep into the apartments of ten men who we know are married, without knowing which apartment belongs to which man, and observe that one of the apartments is as messy as a bachelor pad. We cannot express with (i) the proposition that it appears that an actually married man is a messy bachelor, as predicted if (32) applies.

(i) # A married bachelor appears/seems messy.

ambiguous between a 'low' reading in which the modifier is interpreted under RC-internal material, and a 'high' one in which it is interpreted RC-externally: If the books Sam attributes to Tolstoy are *Anna Karenina* (AK) and *War and Peace* (WP), (36) can have both a true and a false reading. Since WP is, in fact, longer than AK, the *high* reading in (36a) is false. But if Sam (wrongly) claimed AK is longer, the *low* reading in (36b) is true.

(36) The longest book that Sam said Tolstoy wrote is Anna Karenina.

a. '*high-reading*': Of the books Sam said Tolstoy wrote, Anna Karenina is the longest.

b. '*low-reading*': Sam said the following: 'Anna Karenina is the longest book Tolstoy wrote'.

Bhatt attributes this ambiguity to reconstruction as well, arguing that the low-reading is obtained when the superlative RC-head reconstructs into the scope of the RC-internal attitude, while the high-reading is obtained when it is interpreted RC-externally. However, the derivation of the low-reading by means of reconstruction is the subject of much controversy in the literature; there is some doubt whether reconstruction should be appealed to in the first place, and within the pro-reconstruction camp, the structure of the low-reading LF is up for debate.[22] Even without committing to a particular position in these debates, our reductive approach to R4-M makes a non-trivial prediction with respect to Bhatt's ambiguity. This is because, while we may argue whether the low-reading *necessitates* reconstruction, there seems to be a consensus that the high-reading *disallows* it. If this is correct, then under the hypothesis we introduce above regarding partial reconstruction, we predict R4-M to be ruled out when a high-reading is *forced*.

The scenarios that render the high and low readings of (36) true differ in whether the embedded subject is taken to have made any claims about the *lengths* of Tolstoy's books. The high-reading can be true even if Sam has made no such claims whatsoever, whereas its low-reading is only true if Sam made a particular claim; namely, that Anna Karenina is the longest of Tolstoy's books. Thus, in the LF for the *high* reading, the superlative operator and the adjective itself must be interpreted outside the intensional environment created by the propositional attitude. Otherwise, the truth of the sentence would hinge on Sam having ascribed particular lengths to those books she attributes to Tolstoy. In other words, the high-reading is incompatible with reconstruction, at least of the superlative and its adjective. Now

22 Heycock (2005) argues against the reconstruction account, suggesting that the low-reading is restricted in ways it cannot explain. Bhatt & Sharvit (2005), Hulsey & Sauerland (2006) and Sharvit (2007) offer counterarguments, but differ in their proposed reconstruction analyses. Charnavel (2022, 2024, 2026) provides a new perspective on the debate, as well as a useful summary of the pros and cons of reconstruction.

we can appeal again to the hypothesis that interpretation of the adjective in an RC-external position necessitates interpretation of the head noun in that same position, to formulate a prediction regarding the availability of R4-M. If the high reading prohibits reconstruction of the adjective, it should be incompatible with R4-M.

Bhatt provides us with a tool to disambiguate the two readings. In particular, he observes that a *negative polarity item* (NPI) intervening between the RC-internal attitude and the modified NP-head has the following effect: an NPI (like *ever*) above the attitude *necessitates* a high-reading of the sentence (37a), and *blocks* it when it is below the attitude (37b).

(37) a. The longest book that Sam **ever** said Tolstoy wrote is AK.

b. The longest book that Sam said Tolstoy **ever** wrote is AK.

What exactly these facts are supposed to teach us is, again, subject to some debate; particularly regarding the unavailability of the high-reading in (37b).[23] However, we can once again shy away from the controversy in testing our prediction *vis-a-vis* R4-M and NPI placement: Given that a high *ever* forces a high-reading, which in turn, disallows reconstruction of the head noun, a high *ever* should block R4-M.

That this is indeed the case can be illustrated with the contrast in (38).

(38) a. The richest judge Sam **ever** said we could face in court is the Libertarian.

b. The richest judge Sam said we could **ever** face in court is the Libertarian.

The NPI in (38a) forces a high-reading of the superlative operator and its adjective in the RC-head, and also appears to render unavailable the R4-M of the sentence: (38a) seems unable to convey that the richest individual Sam ever said we could face as the judge in court is the libertarian. This is exemplified by the fact that the sentence seems false in a scenario like (39), which involves no actual judges and thus verifies only its R4-M. On the other hand, when the NPI is low as in (38b), the sentence seems true in such an exclusively-R4-M-verifying scenario.[24]

23 See in particular the exchange in Bhatt 2002, Bhatt & Sharvit 2005, Heycock 2005, 2019. Bhatt, as mentioned above, attributes the high/low ambiguity to reconstruction. He thus explains the NPI facts as follows: Assuming NPIs are licensed only in the scope of downward-entailing operators (Ladusaw 1979), and that the superlative is the only downward-entailing operator in (37), the low-reading is blocked for (37a) because it requires reconstruction of the superlative to a position below *ever*, leaving the NPI unlicensed. In (37b), *ever* is in the scope of the superlative even after reconstruction, licensing the low-reading.

24 We have so far intentionally remained agnostic about the relation between the low-reading and reconstruction. Note for completeness, however, that since we take reconstruction to be a necessary ingredient of R4-M, and we observe that a sentence like (38b) is felicitous in exclusively R4-M-verifying scenarios, we can no longer remain fully agnostic with respect to this issue. If R4-M is parasitic on reconstruction, than we are committed to the low-reading being derivable in principle from an LF with a reconstructed RC-head. This is a different claim from the one made in Bhatt

(39) **Context:** As in (35a), Alex is facing trial for stealing from the rich to give to the poor. She again wants her lawyer to connect in person with judicial candidates (none of whom is or has ever been a judge), but is now unsure which of the candidates stands a chance to win, and which of them are too rich to ever be sympathetic to her cause. To avoid wasting time on hopeless candidates, she asks Sam, a consultant, for guidance. Sam suggests that only a Libertarian and two Democrats are realistic candidates, noting that the libertarian is the richest among them. Alex takes Sam to be an authority on these matters, and accepts her word as the truth.

This illustrates that the high-reading is incompatible with R4-M, which again reinforces the correlation between R4-M and reconstruction, on the assumption that the high-reading itself is incompatible with reconstruction.[25]

6 Potential overgeneration

Above, we advance the claim that R4-M is restricted to reconstructed RCs. Next, we address a potential overgeneration challenge due to an anonymous reviewer, suggesting that our account, while less permissive than Szabó's (2010, 2011) bare-quantification approach, is still *too permissive*; there are structures in which reconstruction is presumably permitted, which nevertheless do not give rise to R4-M. These involve RCs whose head-NPs are not bare nouns, but rather modified by certain PPs or by multiple RCs, which seem to resist the reconstruction required to achieve R4-M. The response we offer will not be definitive. Rather, we will attempt to sketch the facts, as judgments are quite delicate, and then contemplate about what might account for the complex empirical picture.

2002, according to which the low-reading necessitates reconstruction, and a less controversial one at that (cf. Heycock 2019).

25 This concludes our illustration of the correlation between reconstruction and R4-M. We would be remiss, however, if we do not comment on the debate about the nature of reconstruction, and how the current discussion interacts with it. The *syntactic* account of reconstruction that we adopt in (17) contrasts with its treatment as a *semantic* process, consisting of type-shifting and abstraction over higher-type variables (e.g., Sharvit 1999). In fact, Grosu & Krifka (2007) defend a semantic account of reconstruction in exactly the RCs discussed here, whose head is interpreted opaquely relative to a RC-internal intensional operator. Our approach to R4-M does not bear directly on this issue. That is, to the extent that semantic reconstruction accounts for the way RCs pattern *vis-a-vis* binding (15a), idioms (15b), resumption (22-26), extraposition (29), and adjectival modification (36), what we illustrate above is merely a correlation between R4-M and reconstruction, *whatever it is taken to be*. Of course, for those who believe the semantic approach does not aptly capture the empirical landscape, this paper provides further support for syntactic reconstruction. We cannot do justice to this issue here, referring the reader to the exchange in Grosu & Krifka 2007, 2022, Bassi & Rasin 2018 for a recent reincarnation of the old debate, with a focus on intensional RCs.

Consider the context in (40a), which verifies, exclusively the R4-M of the sentences below. Given that on the other side of the courtroom door there is but one judge waiting to face Alex, the paper handed to her cannot possibly list six such judges. Rather, it lists six individuals who each could be that one judge. The sentence in (40b) seems true in this context, as it seems to have a R4-M paraphrasable as in (40e). The sentences in (40c-40d), on the other hand, seem false, suggesting they lack that reading.

(40) a. **Context:** A judicial election has occurred; of ten candidates, none of whom were previously judges, one prevailed and filled a vacancy on the court. Alex will face that winner in an upcoming trial, but was not informed on the outcome ahead of time. As she sits outside the courtroom with her she wants to refresh her memory regarding who she might face. Her lawyer does not remember the exhaustive list of candidates, but lists those he remembers on a piece of paper, which he hands to Alex, saying:

b. I listed here six judges we could face today on the other side of this door. **true** in (40a)

c. I listed here six judges on the other side of this door that we could face today. **false** in (40a)

d. I listed here six judges that are on the other side of this door who we could face today. **false** in (40a)

e. I listed here six individuals who we could face as judges on the other side of this door today. **true** in (40a)

This judgment is unexpected on our approach to R4-M. In each of (40c-40d), there is an additional element modifying the noun *judges* besides the modal-embedding RC that this noun heads: a PP in (40c) and a second RC in (40d). If these additional modifiers can be merged in the gap position of the intensional RC as schematized in (41)—and nothing in our system currently suggests that they cannot—then they should be able to reconstruct into that position, below the modal, and the sentence should have a true reading, meaning roughly what (40e) does.

(41) $[\text{D}\ [[_{\text{NP}}\ \text{N}\ [\textbf{modifier}]]][_{\textbf{RC}}\ \ldots\ \text{modal}\ \ldots\ [_{\text{NP}}\ \text{N}\ [\textbf{modifier}]]]]]$

A response to this challenge might consist of positing a syntactic principle that rules out structures like (41). Perhaps, for instance, structures like (41) are either ruled out, or heavily dispreferred relative to an alternative structure, in which the additional adjuncts modifying the head-NP do not merge RC-internally and raise with the head-NP. Rather, the head-NP merges RC-internally and raises on its own, and additional adjuncts are merged countercyclically, only to the copy of the NP in its derived position. This will prevent these modifiers from ever reconstructing into a position below the RC-internal modal, blocking R4-M.

(42) a. [D [[$_{\text{NP}}$ N][**RC** … modal … [$_{\text{NP}}$ N]]]] (raising of head-NP)

b. [D [[$_{\text{NP}}$ N [**modifier**]][**RC** … modal … [$_{\text{NP}}$ N]]]] (late-merge)

If this account is to be pursued, at least two questions must be addressed (though, alas, we are not currently in a position to answer either of them). One is empirical. Is there a *bias* against derivations like (41), or a *constraint* that categorically rules them out? While the sentences in (40c-40d) certainly seem categorically false in their respective exclusively-R4-M-verifying scenario, we can come up with cases that are not as clear. For instance, consider (43b) in the exclusively-R4-M-verifying scenario in (43a).

(43) a. **Context:** A linguistics department is hiring a new faculty member. Alex, a faculty-member who is not part of the search committee, has particular preferences: She would like to hire a graduate of the University of California. After forming a long-list consisting of ten candidates, all of whom are currently finishing up their graduate studies, the head of the committee reaches out to Alex, saying: You'll be happy to hear that…

b. …there are four professors with UC dissertations we could be welcoming to our department in the fall.

The judgments we have collected regarding (43) vary: almost all speakers we have consulted with found it harder to accept than our simpler examples of R4-M, where the head-NP is bare, but quite a few did. So the question is what sets the adjunct in (43b) apart from the ones in (40c-40d), and more generally, adjuncts that do not categorically block R4-M from those that do.

The second question is theoretical. If indeed there is (at the very least) a strong preference for derivations like (42) over (41), why should this be the case? One conjecture is that the tendency to merge things where we hear them is intended to ease the job of the linearization algorithm, which faces less choice points when linearizing a structure like (42) than it does given a structure like (41); only in the latter structure are there two copies of a modifier that needs to be linearized in just one position. We leave further discussion for future research.

7 Conclusion

Semantic orthodoxy has it that the quantificational scope of a determiner is not fully divorced from the intensional status of its restrictor. Szabó (2010, 2011) resurrects Fodor's (1970) old claim that it is, introducing a particular kind of intensional sentence judged true in a scenario that verifies it only if some determiner it contains is taken to quantify unrestrictedly, with the NP "restrictor" (a misnomer on this

approach) evaluated in the domain of an intensional operator in the determiner's scope. Szabó uses this data, exemplifying a reading that we call here R4, to motivate an unorthodox "bare quantification" strategy for natural language, which splits determiners from their restrictors at LF, allowing them to quantify unrestrictedly over the entire domain of quantification. Surprisingly, Szabó's data has not received much attention in the linguistic literature, despite its potential far-reaching implications for mainstream theories of intensionality and scope, which rule out R4.

We attempt to fill the gap by initiating a defense of orthodoxy, so to speak. First, we challenge the leap from the data to the conclusion by showing that bare quantification not only overgenerates unattested R4s, but also undergenerates a particular subclass of cases involving modal operators that Szabó offers as an argument in its favor. We then show that the undergenerated cases can be derived from a construction already independently argued to be available in natural language, of reconstructed intensional RCs, in which quantification is not bare, but restricted. The reconstruction account makes testable predictions that are indeed corroborated.

The scope of the challenge posed by Szabó is thus diminished, but the challenge itself is not fully defused. This is because—as mentioned at the outset—not *all* examples offered in favor of bare quantification are amenable to the reduction we defend for the modal cases. To see this, consider (44-45). Example (44) demonstrates the existence of R4 relative to an attitude predicate: Since Alex did not count her allegations, she does not believe *de dicto* that eleven thieves live in her building, and since the officer need not believe Alex's allegations, (44b) is not a statement regarding Alex's beliefs about eleven *actual* thieves, ruling out its *de re* and *third* readings. The sentence is thus true only if it expresses what the R4 paraphrase in (44c) does.

(44) a. **Context:** Under questioning by the police, Alex is presented with photos of her neighbors, and identifies those she thinks are thieves. She does not count her allegations, but an officer does, and reports to a colleague:

b. Alex believes that eleven thieves live in her building.

c. There are eleven individuals in the actual world such that Alex believes each of them to be thieves living in her building.

(adapted from Szabó 2010: ex. 7)

In (45), it is matrix past tense that creates the intensional environment relative to which R4 is observed: In the context of (45a), the tour-guide does not intend to convey the *de dicto* interpretation of (45b), on which there is a past interval t such that fifteen high-schoolers *at* t previously attended public elementary schools. Nor is she saying anything about fifteen *utterance-time* high-schoolers, given that the camp shut down and its attendees are no longer in their teens, thus ruling out the *de*

re and *third* readings. Rather, she means to convey what (45c) does; i.e., the R4 of (45b).

(45) a. **Context:** We are on a guided tour of what was once a prestigious camp for teenagers that had shut down years ago. The camp catered to teenagers educated in affluent private elementary schools. Every once in a while it also accepted students from the public school system, though never more than a handful at any given point in its existence. The tour guide says: this camp served some of the richest families in the state, but...

b. There were (also) fifteen high-schoolers/teenagers here who attended a public elementary school.

c. There are fifteen individuals, such that for each there is a past time interval during which they were high-schoolers/teenagers and another past time interval during which they attended a public elementary school.

(our example, modeled after Szabó 2011: ex. 28)

But (44b) contains no RC at all, and so the reconstruction account we propose for the modal cases is simply irrelevant. And while (45b) does contain relativization, its ability to convey the R4 paraphrase in (45c) cannot be due to reconstruction: Since one cannot be a teenager and attend an elementary school at the same time, reconstruction of the head-NP in (45b) results in interpreting it contemporaneously with a predicate that is temporally-incompatible with it, in the scope of embedded past tense. Alternative mechanisms are thus required to derive R4 in these examples.

We must leave any exploration of these alternative mechanisms to future research. Our argument from overgeneration against the bare quantification strategy, however, clearly suggest that whatever they are, these mechanisms must be non-trivially restricted by—one would hope—the particular features of the intensional environments involved. In fact, recent proposals in the literature already point in exactly that direction. In particular, it has been suggested that the R4 of belief reports like (44b) can be derived as particular instances of another phenomenon, first observed by Fauconnier (1984), and termed *revisionist belief* by Blumberg & Lederman (2021), in which an agent is truthfully ascribed belief in a proposition p despite their belief state failing to entail p, as long as p is entailed by a revision of their beliefs in light of the reporter's presuppositions.[26] Furthermore, in Benbaji-Elhadad 2026b, we offer an account of the temporal cases that does not appeal to bare quantification. Here, we hope to have convinced the reader with the

26 Cf. Benbaji-Elhadad 2023. Mayr & Schmitt (2026) defend this approach by illustrating that R4 relative to attitudes seems subject to the same constraints governing the distribution of revisionist belief. However, in Benbaji-Elhadad 2026a, we point to some potential challenges their argument faces.

argument above that one is better off attempting to derive different instances of R4 from distinct construction-specific mechanisms, than to unify them under a single unrestricted one.

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